

and Python. TensorFlow was developed by Google for better AI applications. It was initially launched to be used in Gmail, Google photos and Google Search. This open source AI tool is also being used for numerical computations by programmers. The easy-to-use interface and architecture of TensorFlow allows you to use its different platforms. There is no restriction regarding its compatibility. You can use it on TPUs, CPUs and GPUs. You can even use it on your PC, mobile and laptop. This tool is effectively helping researchers in their numerical computation and applications in neural networks. TensorFlow was launched under the Apache 2.0 open source licence in 2015.

Link: <https://www.tensorflow.org/>

Deeplearning4j: As the name suggests, Deeplearning4j is an open source Java AI tool specially designed for deep learning. It was released under the Apache License 2.0, having been created by a group of AI researchers based in San Francisco and Tokyo.

It makes use of its own deep learning library created for Java Virtual Machine (JVM). This library is called ND4J and it works with both CPUs as well as GPUS. The framework is used for forming neural nets and has many advanced visualisation tools. This tool has many diverse academic applications, and is being used in the fields of cyber security and image recognition. It has been integrated with other open source AI platforms like Keta and TensorFlow. It can make use of different API languages like Python and Clojure.

Link: <https://deeplearning4j.org/>

Caffe: This is an open source AI tool developed by the Berkeley Vision and Learning Center. It encourages deep learning and is able to process millions of images per day. It has remarkable speed and was released under the BSD 2-Clause licence. This dynamic AI tool has been written in C++ and has a Python interface. It comes with an impressive architecture that allows you to switch between the CPU and GPU. This AI tool is being used by academic researchers in various projects and has even found large scale application in the multimedia and vision domains.

Link: <http://caffe.berkeleyvision.org/>

ONNX (Open Neural Network Exchange): ONNX is an open source AI format which is extensively used by AI researchers in deep learning models. It is primarily a Facebook open source project but is also endorsed by Microsoft and AWS. This open source ecosystem for deep learning was developed in 2017 when both the tech giants, Facebook and Microsoft, came together to create a system for switching ML frameworks.

Microsoft contributed by adding its Cognitive Toolkit and Project Brainwave platform to the initiative. This platform

allows its users to apply its innovative contents like extensible computation graph models, and it allows them to change their networks according to their requirements.

Link: <https://onnx.ai/>

Oryx 2: This is an open source framework built on Spark and Apache Kafka. Oryx 2 allows for real-time machine learning on a bigger scale. This platform is being used by AI researchers to build applications, and for classification and clustering.

Link: <http://oryx.io/>

Apache SystemML: This is a flexible open source AI platform that focuses on Big Data and has been designed for complex mathematical problems. It can scale to Spark and Hadoop. It runs on an R Python-like syntax. It finds application in deep learning with GPUs, and in neural network architectures for training.

Link: <https://systemml.apache.org/>

MLlib: This was launched by Apache Spark and is a machine learning library for learning algorithms. This AI tool uses different widely used programming languages like R, Scala, Java and Python. It can run on many different platforms including Hadoop, Kubernetes or in the cloud. This library contains many deep learning and core ML algorithms. It makes ML easier and more practical for AI researchers.

Link: <https://spark.apache.org/mllib/>

A final word

The use of the above AI tools is revolutionising industries immensely. It is being predicted that everything that has been invented until today can be reinvented in a much better and more efficient way in the next 15 years, with the help of AI.

Businesses are now being revolutionised with CEOs engaging with AI as well. AI no doubt offers you the most noteworthy and extraordinary possibilities to expand your business and harness the untapped potential of modern technology.

There is vast potential to use these open source AI tools in healthcare, robotics, food and financial trading. Unfaltering success is expected if businesses start keeping up with these advances in AI in health, transport, retail, banking and other domains. What we've experienced of AI till now is just the tip of the iceberg! **END** 🐧

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